Towards a Digital Biomarker Ontology: A Heart Failure Health Markers Application

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1 Introduction

The increasing use of wearable sensors and home monitoring devices creates a continuous stream of biomarkers such as heart rate, blood pressure, activity, or sleep. When linked to clinical outcomes, these biomarkers can provide valuable insights into disease progression, treatment adherence, and recovery trajectories. However, a major obstacle to realising this potential is the heterogeneity of data structures and semantics across devices, vendors, and healthcare systems. The **Digital Biomarker Ontology (DBO)** addresses this gap by introducing a semantic layer that aligns observation data with clinically meaningful concepts.

2 Ontology design

The Digital Biomarker Ontology (DBO) is implemented as a modular ontology, reusing external vocabularies and ontologies wherever possible. Its design followed the SABiO and the Linked Open Terms engineering methodologies, emphasising the usage of competency questions (CQs) and the reuse of existing standards. The design phase consisted of both a top-down and a bottom-up phase: the competency questions questions were derived from the *LUMC The Box* remote monitoring program. Additionally, the available data informed the ontology design. We give a selection of the CQs:

- **Competency questions atomic level:** 1. What is the value of this measurement? 2. When was the measurement made? 3. With which device was the measurement made?
- **Competency questions connectivity level:** 1. Which measurements were recorded between two consecutive hospital visits? 2. Did the patient perform at least n weight measurements per week? 3.What were the lowest|highest|average value per property, such as weight, blood pressure, or glucose?
- **Competency questions use-case level:** 1. To which extent did the patient follow the measurement protocal?, 2. What is the difference between the hospital measurements and the wearable measurements?

The DBO (figure 1) reuses the following ontologies and vocabularies:

1. **SAREF** and its extensions **SAREF4WEAR** and **SAREF4EHAW** provide core sensor and observation concepts. The SAREF ontology has been modelled following the SSN/SOSA ontology and the OGC O&M standard.
2. The **QUDT Units** vocabulary provides URIs for all units of measure.
3. **LOINC** and **SNOMED CT** provide URIs for medical terminology.
4. **Friend of a Friend** provides concepts and properties for patients and some of their personal information.

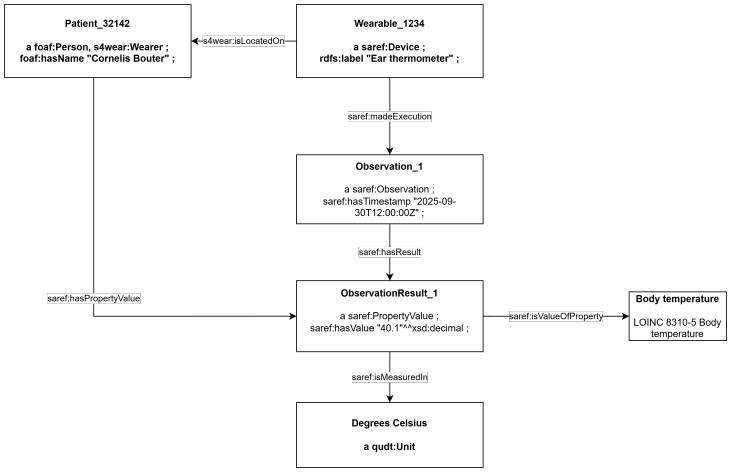

Fig. 1. Example of a measurement representation in DBO showing the reused concepts and properties resources from SAREF and by extension SOSA/SSN, SAREF4WEAR, LOINC, and QUDT.

3 Evaluation: validation and verification

We **verified** the ontology by translating all natural language CQs to SPARQL queries that return the required answers. The ontology design is **validated** by transforming the LUMC source data to the ontology model. The ontology is exemplified by a synthetic data knowledge graph.

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